Topic: Laplace transforms of step functions

Question: Find the Laplace transform of the function $g(t)$.

$$g(t) = 5u_1(t) - 9u_3(t)$$

Answer choices:

A $\mathscr{L}(g(t)) = \dfrac{5e^{-s}}{s} - \dfrac{9e^{-3s}}{s}$

B $\mathscr{L}(g(t)) = \dfrac{e^{-s}}{s} - \dfrac{e^{-3s}}{s}$

C $\mathscr{L}(g(t)) = \dfrac{5e^{-s}}{s} + \dfrac{9e^{-3s}}{s}$

D $\mathscr{L}(g(t)) = \dfrac{e^{-s}}{s} + \dfrac{e^{-3s}}{s}$

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Solution: A

We know that the Laplace transform of $u_c(t)$ is given by

$$\mathscr{L}(u_c(t)) = \frac{e^{-sc}}{s}$$

Therefore, the Laplace transform of $5u_1(t)$ must be

$$\frac{5e^{-s}}{s}$$

and the Laplace transform of $-9u_3(t)$ must be

$$-\frac{9e^{-3s}}{s}$$

Putting these together, the Laplace transform is

$$\mathscr{L}(g(t)) = \frac{5e^{-s}}{s} - \frac{9e^{-3s}}{s}$$

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Topic: Laplace transforms of step functions

Question: Find the Laplace transform of the function $f(t)$.

$$f(t) = \begin{cases} 2 & 0 \leq x < 1 \\ 5 & 1 \leq x < 3 \\ -1 & x \geq 3 \end{cases}$$

Answer choices:

A $\mathscr{L}(f(t)) = 2 + 5e^{-s} - e^{-3s}$

B $\mathscr{L}(f(t)) = 2 + \dfrac{3e^{-s}}{s} - \dfrac{6e^{-3s}}{s}$

C $\mathscr{L}(f(t)) = \dfrac{2}{s} + \dfrac{5e^{-s}}{s} - \dfrac{e^{-3s}}{s}$

D $\mathscr{L}(f(t)) = \dfrac{2}{s} + \dfrac{3e^{-s}}{s} - \dfrac{6e^{-3s}}{s}$

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Solution: D

Remember that a Heaviside function represents a switch, so we need one Heaviside function every time we make a switch in the value of $f(t)$. There are two switches, one from 2 to 5, and one from 5 to -1, so we'll need two Heaviside functions in order to express $f(t)$.

We write a Heaviside function as $u_c(t)$, where the switch occurs at $t = c$. The switches in this function $f(t)$ are at $t = 1$ and $t = 3$, which means the two Heaviside functions we'll need are $u_1(t)$, and $u_3(t)$.

When the first switch turns on at $t = 1$, the value of the function immediately shifts from 2 to 5 ($5 - 2 = 3$), so we can write

$$f(t) \approx 2 + 3u_1(t)$$

When the second switch turns on at $t = 5$, the value of the function immediately shifts from 5 to -1 ($-1 - 5 = -6$), so we can write

$$f(t) \approx 2 + 3u_1(t) - 6u_3(t)$$

We know that the Laplace transform of $u_c(t)$ is given by

$$\mathcal{L}(u_c(t)) = \frac{e^{-sc}}{s}$$

Therefore, the Laplace transform of $3u_1(t)$ must be

$$\frac{3e^{-s}}{s}$$

and the Laplace transform of $-6u_3(t)$ must be

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$$-\frac{6e^{-3s}}{s}$$

Putting these together, the Laplace transform is

$$\mathscr{L}(f(t)) = \frac{2}{s} + \frac{3e^{-s}}{s} - \frac{6e^{-3s}}{s}$$

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Topic: Laplace transforms of step functions

Question: Find the inverse Laplace transform.

$$G(s) = \frac{24e^{-5s}}{s^5}$$

Answer choices:

A $\mathscr{L}^{-1}(G(s)) = t^4u(t-5)$

B $\mathscr{L}^{-1}(G(s)) = (t-5)^4u(t-5)$

C $\mathscr{L}^{-1}(G(s)) = (t-5)^4$

D $\mathscr{L}^{-1}(G(s)) = (t-5)^4u(t)$

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Solution: B

We'll start by pulling e^{-5s} out of the fraction, so that the function is then in the form $G(s) = e^{-5s}F(s)$.

$$G(s) = \frac{24e^{-5s}}{s^5}$$

$$G(s) = e^{-5s}\left(\frac{24}{s^5}\right)$$

$$F(s) = \frac{24}{s^5}$$

The inverse transform of $F(s)$ is

$$\mathscr{L}^{-1}(F(s)) = \mathscr{L}^{-1}\left(\frac{4!}{s^{4+1}}\right) = t^4$$

Which means the inverse transform of $G(s)$ is

$$\mathscr{L}^{-1}(G(s)) = u_5(t)f(t-5)$$

$$\mathscr{L}^{-1}(G(s)) = (t-5)^4 u(t-5)$$

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